

in a conventional car. At temperatures above 300°C it moved around too fast for the chemists to keep track of it!

Lithium-ion Battery (2007)

Angela Belcher and colleagues at MIT built a lithium-ion battery with a common type of virus that is non-harmful to humans, using a low-cost and environmentally benign process. The batteries have the same energy capacity and power performance as state-of-the-art rechargeable batteries being considered to power plug-in hybrid cars, and they could also be used to power personal electronic devices.

3D DNA Structures (2009–2010)

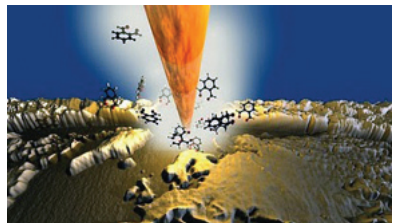
Nadrian Seeman and colleagues at New York University created several DNA-like robotic nanoscale assembly devices. One is a process for creating 3D DNA structures using synthetic sequences of DNA crystals that can be programmed to self-assemble using “sticky ends” and placement in a set order and orientation. Nanoelectronics could benefit: the flexibility and density that 3D nanoscale components allow could enable assembly of parts that are smaller, more complex, and more closely spaced.



Inventor of DNA-like robotic nanoscale assembly devices

Smallest Relief Map (2010)

IBM used a silicon tip measuring only a few nanos at its apex (similar to the tips used in atomic force microscopes) to chisel away material from a substrate to create a complete nanoscale 3D relief map of the world one-one-thousandth the size of a grain of salt—in 2 minutes and 23 seconds. This activity demonstrated a powerful patterning methodology for generating nanoscale patterns and structures as small as 15 nanos at greatly reduced cost and complexity, opening up new prospects for fields such as electronics, optoelectronics, and medicine. 



World's Smallest Relief Map